RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

Tutorial S5: Hypothesis Testing

Section A (Basic Questions)

1 'Brilliant' fireworks are intended to burn for 40 seconds. A random sample of 50 'Brilliant' fireworks is taken. Each firework in the sample is ignited and the burning time, x seconds, is measured. The results are summarised by $\sum(x-40)=-27$, $\sum(x-40)^2=167$.

Test, at the 5% level of significance, whether the mean burning time of 'Brilliant' fireworks differs from 40 seconds. [7]

State what you understand by the expression 'at the 5% significance level' in the context of this question. [1]

State with a reason, whether, in using the above test, it is necessary to assume that the burning times of 'Brilliant' fireworks have a normal distribution. [1]

[p-value = 0.0304 ≤ 0.05, reject H_0]

9205/1999/02/Q10(b) (modified)

Solution:

Let $y=x-40$, then $\bar{y}=\bar{x}-40$, $\sum y=-27$ and $\sum y^2=167$

From the sample, $\bar{x}=\frac{-27}{50}+40=39.46$, ${s_x}^2={s_y}^2=\frac{1}{49}\left[167-\frac{(-27)^2}{50}\right]=3.1106$ (5sf)

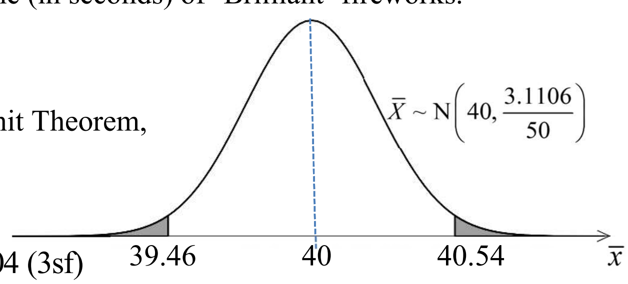
Let μ denote the **population mean** burning time (in seconds) of 'Brilliant' fireworks.

$H_0: \mu=40$ vs $H_1: \mu\neq 40$

Perform a 2-tail test at 5% significance level.

Under H_0, since n = 50 is large, by Central Limit Theorem,

$\bar{X}\sim N\left(40, \frac{3.1106}{50}\right)$ approximately.



Using a z-test, p-value $=2P\left(\bar{X}\leq 39.46\right)=0.0304$ (3sf)

Since p-value $=0.0304\leq 0.05$, we **reject** H_0 and conclude that there is **sufficient** evidence, at the 5% significance level, that the <u>mean burning time of 'Brilliant' fireworks differs from 40 seconds</u>.

This underlined statement always refers to H_1.

'at the 5% significance level' means that there is a probability of 0.05 of wrongly concluding that the mean burning time of 'Brilliant' fireworks differs from 40 seconds when it is in fact 40 seconds.

It is not necessary to assume that the burning times of 'Brilliant' fireworks have a normal distribution. Since the sample size 50 is large, by Central Limit Theorem, the <u>sample mean</u> $\bar{X}$ follows a normal distribution approximately.

2 An office worker, Natalie, has diabetes and has to monitor her blood glucose levels, which vary throughout the day. The results from a sample of 75 readings, x (in mmol/L), taken at random times over a week, are summarised by $\sum x = 511.5$ and $\sum x^2 = 4027.89$.

(i) Calculate unbiased estimates of the population mean and variance for the blood glucose levels. [2]

(ii) Test at 5% significance level whether Natalie's mean blood glucose level, μ (in mmol/L), is greater than 6.0. You should state your hypotheses and give your conclusion in context. [4]

(iii) State, giving a reason, whether the conclusion of the test in part **(ii)** would be valid if the 75 readings were all taken at weekends. [1]

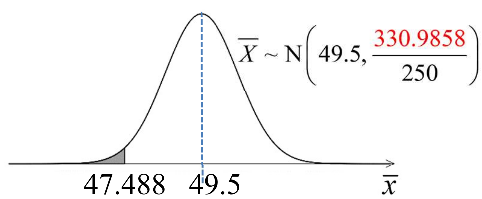
(iv) Explain why there is no need for Natalie to know anything about the population distribution of the glucose blood levels when carrying out the test in **(ii)**. [1]

[**(i)** $\bar{x} = 6.82; s^2 = 7.29$, **(ii)** p-value = 0.00427, reject H_0]

NYJC/9758/2021/02/Q10 (modified)	
Solution:	
(i)	Unbiased estimates for population mean and population variance: $\bar{x} = \frac{511.5}{75} = 6.82$ $s^2 = \frac{1}{74}\left(4027.89 - \frac{511.5^2}{75}\right) = 7.29$
(ii)	$H_0 : \mu = 6.0$ vs $H_1 : \mu > 6.0$ Perform a 1-tail test at 5% significance level Under H_0, since sample size of 75 is large, by Central Limit Theorem, $\bar{X} \sim N\left(6, \frac{7.29}{75}\right)$ approximately. Using a z-test, p-value $= P\left(\bar{X} \geq 6.82\right) = 0.00427$ (3sf) Since p-value $= 0.00427 \leq 0.05$, we **reject** H_0 and conclude that there is **sufficient** evidence, at 5% significance level, that Natalie's mean blood glucose level is greater than 6.0. This underlined statement always refers to H_1.
(iii)	Readings at weekend may be biased by different lifestyle, so results may not be valid. Recall that the sample used for hypothesis testing must be a random sample.
(iv)	Since the sample size 75 is large, by Central Limit Theorem, the sample mean $\bar{X}$ follows a normal distribution approximately. Therefore, there is no need to know anything about the population distribution of the glucose blood levels.

3 At an early stage in analyzing the marks scored by the large number of candidates in an examination paper, the Examining Board takes a random sample of 250 candidates and find that the marks, x, of these candidates give $\sum x = 11\,872$ and $\sum x^2 = 646\,193$. Using the figures obtained in this sample, the null hypothesis $\mu = 49.5$ is tested against the alternative hypothesis $\mu < 49.5$ at the α% significance level. Determine the set of values of α for which the null hypothesis is rejected in favour of the alternative hypothesis.

[$\{\alpha \in \mathbb{R} : 4.02 \le \alpha \le 100\}$]

9205/N1988/02/Q10 (modified)
Solution:
From the sample, $\bar{x} = \dfrac{11872}{250} = 47.488$, $s^2 = \dfrac{1}{249}\left(646\,193 - \dfrac{11\,872^2}{250}\right) = 330.9858$. $H_0: \mu = 49.5$ vs $H_1: \mu < 49.5$ Under H_0, since $n = 250$ is large, by Central Limit Theorem, $\bar{X} \sim N\left(49.5, \dfrac{330.9858}{250}\right)$ approximately. Using a z-test, p-value $= P\left(\bar{X} \le 47.488\right) = 0.0402$ (3sf). If H_0 is rejected at α% significance level, p-value ≤ significance level $0.0402 \le \dfrac{\alpha}{100}$ $\alpha \ge 4.02$ The set of values of α is [4.02, 100] or $\{\alpha \in \mathbb{R} : 4.02 \le \alpha \le 100\}$.  Note: The significance level cannot be more than 100%